

18.8.26

DAY - 13

DEBUGGING :-

Debugging Steps :

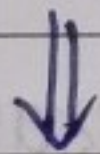
STEP - 1 : DESCRIBE THE PROBLEM

* The first step of solving Problem is being able to describe the problem.

Always :

① When you come across a problem in your code :-

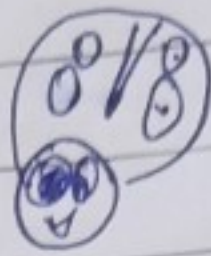
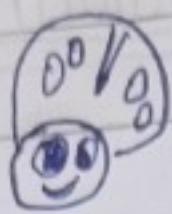
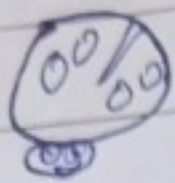
② TRY TO DESCRIBE THE PROBLEM CLEARLY



SO THAT YOU REALLY UNDERSTAND WHAT ISSUE IS?
& WHAT'S GOING ON?

③ And Then ; Test your assumptions

STEP - 2 : NEXT ! REPRODUCE THE BUG



Reproduce the bug means : Makes the same bug happen again intentionally, using same steps or conditions that caused it

Example :

```
age = int(input("Enter your age:"))
```

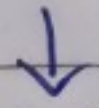
```
print(100/age)
```

① some one Reports : "Program crashes when i enter 0"

Now:

TO REPRODUCE THE BUG YOU DO :

Run - program (Δ)



Enter : 0 (intentionally)



program crashes → Oh! zero division error

NOW YOU DO FIXING;

```
age = int(input("Enter your age:"))
```

```
if age == 0 :
```

```
    print("Age can't be zero")
```

```
else:
```

```
    print(100/age)
```


Example :

```
from random import randint  
dice_images = ["①", "②", "③", "④", "⑤", "⑥"]
```

```
dice_num = randint(1, 6)
```

```
print(dice_images[dice_num])
```

Runs smoothly But
occasionally error
comes. So;

* Let's Reproduce the Bug Q ;

```
dice_num = 1
```

```
print(dice_images[1])
```

Output :

②

Oh! GOT IT List start counting
from 0.

```
dice_num = 6
```

```
print(dice_images[6])
```

Here: 1 2 3 4 5 6
[①, ②, ③, ④, ⑤, ⑥]

↓
out of index

Okay it's the Bug;

∴

```
dice_num = randint(0, 5)
```


Step-3 : Play Computer

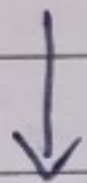
It means you act like the computer and manually execute the code step by step in your head or paper.

Step-4 : Fix the Errors

* THERE ARE MORE TYPE OF ERRORS ;

① Errors which can be pointed out by editor

② Errors which can't be pointed by editor ^{like} logical errors



THIS CASE COPY THE ERROR LINE Paste it into Google Search. (stack overflow)

→ Step-5 : USE PRINT STATEMENT

Step-6 : USE DEBUGGER USE (●) Break-point ; puts a break on Particular line.

SIC
GELA
Step-7 : RUN - OFTEN (Δ)

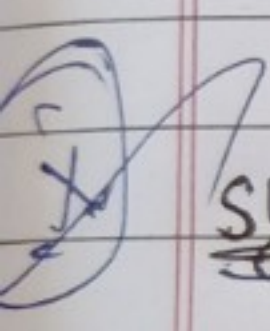
RUN after every little of code execution

I. Introduction to try and except keyword :

try and except are used in Python to handle run-time errors (Exceptions) without crashing the program.

* try → It contains the code that might cause an error.

* except → tells Python what to do if that specific error occurs.



Syntax :

try :
 # code that might cause an error

except <Error Type> :
 # what to do if that error occurs

Example :

```
age = int(input("Enter your age :"))

if age > 18 :
    print("You can drive")
```

OUTPUT :

Enter your age : twelve

VALUE ERROR : - - -

Now:

try :

age = int(input("Enter your age :"))

except Value Error :

print("you have typed invalid number, Please
try again with numerical")

age = int(input("Enter your age :"))

if age > 18 :

print("you can drive")

OUTPUT :

Enter your age : twelve

you have typed invalid number, Please try again with numerical

Enter your age : 20

you can drive

WHAT'S USE OF IT ?

To prevent your program from crashing when a known run-time problem occurs, and handle that problem in a controlled way.

COMPLETED CODING CHALLENGES :

